

Agile Lifecycle

Hospital Appointment Reminder and Scheduling System

Step 1: Features of the System

1. Patient registration and login
2. Manage profile
3. View doctor availability
4. Book appointment
5. Automated appointment confirmation
6. Automatic cancellation logic
7. Doctor's dashboard
8. Reschedule appointment
9. Appointment reminder
10. Appointment reports

Step 2: Prioritize the features

1. Patient Registration and Login
2. Book Appointment
3. View Doctor's Availability
4. Automated Appointment Confirmation
5. Doctor's Dashboard
6. Appointment Reminder
7. Auto-cancellation Logic
8. Reschedule Appointment
9. Appointment Reports
10. Manage Profile

Step 3: Divide the features Into Iterations

- **Iteration 1:** Basic Booking
- **Iteration 2:** Reminders and Automation
- **Iteration 3:** Flexibility and Doctor Access
- **Iteration 4:** Improvements and Reporting

Step 4: Goal, Features, Tasks, Testing Activities, Feedback Source For Each Iteration

Iteration 1: Basic booking

Goal

A patient can register, log in, see when a doctor is free, and book an appointment.

Features

1. Patient registration and login
2. View doctor availability
3. Book appointment

Tasks

1. Design the patient registration and login interface.
2. Create the patient registration and login functionality.
3. Create the doctor availability interface.
4. Add doctor schedules and available time slots.
5. Develop the appointment booking functionality.
6. Connect booking with the database.
7. Ensure booked time slots are updated accordingly.

Testing Activities

1. Test patient registration with valid and invalid details.
2. Test patient login with correct and incorrect credentials.
3. Test whether available doctors and time slots are displayed correctly.
4. Test whether a patient can successfully book an appointment.
5. Test that unavailable time slots cannot be booked.
6. Test that appointment details are correctly saved in the database.

Feedback Source

1. Patients
2. Receptionist

Iteration 2: Reminders and automation (your most important feature)

Goal

The system reminds patients about their appointments, asks them to confirm, and automatically cancels appointments that aren't confirmed or attended.

Features

1. Appointment reminders
2. Automated confirmation request
3. Automated cancellation logic

Tasks

1. Design the appointment reminder process.
2. Develop the appointment reminder functionality.
3. Create the automated confirmation request.
4. Develop the confirmation response process.
5. Develop the automated cancellation logic.
6. Connect reminders and confirmation requests to booked appointments.
7. Configure the system to identify appointments that are not confirmed or attended.

Testing Activities

1. Test whether reminders are sent for upcoming appointments.
2. Test whether patients receive the confirmation request.
3. Test confirmation of an appointment.
4. Test cancellation when an appointment is not confirmed.
5. Test the automatic cancellation logic under different conditions.
6. Test that confirmed appointments are not cancelled incorrectly.

Feedback Source

1. Patients
2. Receptionist

Iteration 3: Flexibility and doctor access

Goal

Patients can move their appointments instead of losing them, and doctors can see their own schedules.

Features

1. Reschedule appointment
2. Doctor's dashboard

Tasks

1. Design the reschedule appointment interface.
2. Develop the appointment rescheduling functionality.
3. Allow patients to select a new available date and time.
4. Update the appointment details after rescheduling.
5. Design the doctor's dashboard.

6. Display doctor appointments and schedules on the dashboard.
7. Connect the dashboard to the appointment database.

Testing Activities

1. Test whether patients can reschedule an existing appointment.
2. Test that only available dates and times can be selected.
3. Test that the old appointment time becomes available after rescheduling.
4. Test that the new appointment details are saved correctly.
5. Test whether doctors can log in and access their dashboard.
6. Test whether the doctor's appointments and schedules are displayed correctly.

Feedback Source

1. Patients
2. Doctors

Iteration 4: Improvements and reporting

Goal

Patients can update their details, and the hospital can generate appointment records and reports.

Features

1. Manage profile
2. Appointment reports

Tasks

1. Design the patient profile management interface.
2. Develop the managed profile functionality.
3. Allow patients to update their personal information.
4. Validate and save updated profile information.
5. Design the appointment reports interface.
6. Develop appointment report generation.
7. Allow the hospital to view appointment records and relevant information.

Testing Activities

1. Test whether patients can access their profiles.
2. Test whether patients can update their details.
3. Test that updated information is correctly saved.
4. Test validation of incorrect or incomplete profile information.

5. Test whether appointment reports are generated correctly.
6. Test whether reports contain accurate appointment information.

Feedback Source

1. Patients
2. Receptionist

WHAT COULD CHANGE IF A TEAM ADOPTS AGILE WITHOUT PROPER PLANNING

If a team adopts Agile without proper planning, the project may become confusing because team members may not know exactly what they are supposed to do. Features may keep changing, some tasks may be forgotten, and team members may end up doing the same work. The project may also take longer because some work might have to be repeated. The quality of the system can also go down if the team rushes to finish tasks. Therefore, even though Agile allows changes, the team still needs a simple plan for each iteration so that everyone knows what they need to do and what they want to achieve.